Sisir Kausthubh

Robotics & Edge-AI Inference Engineer

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SUMMARY

I make robot foundation models fast enough for the real world and frugal enough for battery-powered robots, while keeping nearly the same accuracy. Every result reports three things together, on the same setup, against a published baseline: **speed** (median and p99 latency, startup), **accuracy** (closed-loop task success on a public benchmark) and **efficiency** (energy and memory per decision).

ROBOTICS

SmolVLA fast path on LIBERO · independent research Oct 2026 – present

- **7.7× faster per decision, no retraining:** 101 → 13 ms median, 120 → 16 ms p99 (Hugging Face SmolVLA, RTX 5090 laptop GPU).
- **9.3× less energy per decision** (10.5 → 1.1 J). **Task success 83.00% vs 83.75%** over the same 400 closed-loop episodes; paired McNemar test, no significant change.
- How: whole-model `torch.compile`, 1 denoising step instead of 10, batched camera encoding, and dropping an unused blank camera slot.
- Also tested TensorRT, INT8 W8A8 and FP8. Rejected FP8 vision: faster, but success fell to 80%. Reported trade-off: 71–89 s cold startup (15 s warm).

OPEN SOURCE & COMMUNITY

- **Hugging Face LeRobot:** reviewed and benchmarked PR #4634 (SmolVLA KV cache + compile); measured 2.2× end to end vs 4.2× claimed. A SmolVLA camera-batching change is in progress.
- **Speaker:** lightning talk on the SmolVLA results, NVIDIA Jetson AI Lab community meeting, 10 Nov 2026.

AI AGENTS & GENAI · public repos, each on one laptop GPU, against a published baseline

- **catalog-automation-agent** (LangGraph, MCP server, n8n): auto-tagged 79.2% of 1,610 ImplicitAVE products at 94.6% accuracy; crash test lost 0 items with 0 duplicates; 45 products/s; ₹0.35 per 1,000 products.
- **visual-product-tagging**: fine-tuned 4B vision-language model, 85.3% on ImplicitAVE (GPT-4V: 86.8%); ₹0.67 per 1,000 products.
- **docs-rag-assistant**: page-cited RAG with a local 4-bit Gemma 4 26B; 61.8% on FinanceBench (paper's GPT-4-Turbo RAG: 19%).
- **document-extraction-pipeline**: receipts and GST invoices to checked rows; 84.0 field F1 on CORD (Donut: 83.3 on the same scorer).

NEXT (DESIGNED, STARTING NOV 2026)

- **GR00T fast-path audit:** does NVIDIA's TensorRT/FP8 path for GR00T N1.7 hold up closed loop? All three pillars, on RTX and Jetson.
- **SmolVLA over ROS 2:** the same closed loop through robot middleware, measuring the latency it adds.
- **Fault-tolerant swarms:** kill units and cut links mid-task, measure work lost and recovery time; LLM agents first, then multi-robot failover in Isaac Lab.
- **Distillation:** large robot models into small ones that fit on a Jetson.

SKILLS

Inference & ML: Python, PyTorch, torch.compile, TensorRT, INT8/FP8 quantization, vLLM, LoRA fine-tuning, Hugging Face (LeRobot, Transformers)
Robotics & sim: VLAs (SmolVLA), LIBERO, MuJoCo, reinforcement learning (DDPG, TD3) · learning ROS 2, Isaac Lab, Jetson
Agents & GenAI: LangGraph, MCP, RAG, vision-language models, evaluation with paired statistics
Tools: Linux, Git, GitHub CLI, uv

EDUCATION

BTech, Computer Science and Engineering (AI) · Amrita Vishwa Vidyapeetham 2025
Earlier projects: DDPG warehouse-robot navigation (ROS), Tamil→Telugu speech-to-speech translation, Indic-language NLP.